

1. Student Management System

CO1-AT3

CSA0911

Programming in Java

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```
class Student
{
    Private String name;
    Private int rollNo;
    Private int mark1, m2, m3;
    void acceptDetails()
    {
        input name;
        input rollNo;
        input mark1, m2, m3;
    }
    double calAvg()
    {
        return (mark1 + m2 + m3) / 3;
    }
    void dispRes()
    {
        double avg = calAvg();
        Print name;
        Print rollNo;
        Print avg;
        if (avg >= 50)
            return "Pass";
        else
            return "fail";
    }
}

class Main
{
    main()
    {
        Student s = new Student();
        s.acceptDetails();
        s.dispRes();
    }
}
```

2. Library Book Management

class Book

```
{  
    private String title;  
    private String author;  
    private String ISBN;  
    private boolean isAvl;  
    void setDetails (String t, String a, String i)  
    {  
        title = t;  
        author = a;  
        ISBN = i;  
        isAvl = true;  
    }  
}
```

void issueBook()

```
{  
    if (isAvl)  
    {  
        isAvl = false;  
        return "issued";  
    }  
    else  
        return "Not Avl";  
}
```

void returnBook()

```
{  
    isAvl = true;  
    Print "Book Returned";  
}
```

void displayDetails()

```
{  
    Print title;  
    Print author;  
    Print ISBN;  
    Print isAvl;  
}
```

class Main

```
{  
    Book b1 = new Book();  
    Book b2 = new Book();  
}
```

b1. Set Details ("Ram", "Python", "001");

b2. Set Details ("Laxman", "Java", "002");

b1. IssueBook();

b1. return Book();

b1. displayDetails();

b2. displayDetails();

3. Employee Salary calculation

class Emp

{

private int empId;

private String name;

private int Pay;

void SetDetails (int id, String n, double Pay) {

empId = id;

name = n;

Pay = Pay;

}

double calGross()

{

return Pay + (Pay * 0.20) + (Pay * 0.10);

}

double calNet

{

double gross = calGross();

return gross - (basic Pay * 0.05) - (basic Pay * 0.05);

}

void display Payslip()

{

Print empId;

Print name;

Print calGross();

Print calNet;

}

class Main

{

main

{

Emp e = new Emp();

e.SetDetails (10, "Ram", 1000);

e.display Payslip();

}

4. Shape Area calculator

```
class Shape
{
    void calArea()
    {
    }
}

class Circle extends Shape
{
    void calArea()
    {
        Print (3.14 * 5 * 5);
    }
}

class Rectangle extends Shape
{
    void calArea()
    {
        Print (10 * 4);
    }
}

class Triangle extends Shape
{
    void calArea()
    {
        Print ((8 * 6) / 2);
    }
}

class Main
{
    main
    {
        Shape s;

        s = new Circle();
        s.calArea();

        s = new Rectangle();
        s.calArea();

        s = new Triangle();
        s.calArea();
    }
}
```

5. vehicle Registration System

```
class vehicle
{
    void display()
    {
        Print vehicle ;
    }
}

class TwoWheel extends vehicle
{
    void display()
    {
        Print Two wheel ;
    }
}

class fourwheel extends vehicle
{
    void display()
    {
        Print four wheel ;
    }
}

class Main
{
    main()
    {
        vehicle v ;
        v = new TwoWheeler();
        v.display();
        v = new fourwheel();
        v.display();
    }
}
```

6. Bank Account hierarchy

```
class BankAcc
{
    accno, holder, bal
    Method withdraw (amount)
}

class Saving Account extends Bank Acc
{
    Method withdraw (amount)
    if amount <= Bal Then

```

Balance = Balance - amount

else

Print "Ins Balance"

and statement

class currentAcc extends BankAcc

overdraft = 10000

Method withdraw (amount)

if amount <= bal + overdraft then

balance = balance - amount

else

Print "Limit exceeded"

end

make object s, c

s. withdraw (2000)

c. withdraw (2000)

7. hospital patient Record system

class patient

private name, dia, pres

Method generate Bill()

end

class impatient extends patient

ward, days, charge

Method generate Bill()

total = days * charge

Print total

end

class outpatient extends patient

Method generate Bill()

Print fee

end

create object i, o

i. generate Bill()

o. generate Bill()

```
class TeachingStaff extends Staff
```

```
Method calAllowance()
```

```
allow = salary * 0.10  
end
```

```
class Professor extends TeachingStaff
```

```
Method calAllowance()
```

```
allow = salary * 0.20  
end
```

```
class Lecturer extends TeachingStaff
```

```
Method calAllowance()
```

```
allow = salary * 0.15  
end
```

```
class NonTeachingStaff extends TeachingStaff
```

```
Method calAllowance()
```

```
allow = salary * 0.08  
end
```

```
create object P, e, n
```

```
P. calAllowance()
```

```
e. calAllowance()
```

```
n. calAllowance()
```

```
end.
```

10. Smart Home Device

```
class SmartDevice {
```

```
Method ExecuteCommand()
```

```
Device
```

```
end
```

```
class SmartLight extends SmartDevice
```

```
Method ExecuteCommand()
```

```
Light
```

```
end
```

```
class SmartThermostat extends SmartDevice
```

```
Method ExecuteCommand()
```

```
Thermostat
```

```
end
```

```
class Object l, t, s
```

```
l. ExecuteCommand();
```

```
t. ExecuteCommand();
```

```
s. ExecuteCommand();
```